PhaseEval: Expert Review Eval Environment/Interface for Agentic XRD Analysis

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OSI LLM Hackathon, MIT, October 21 to 22, 2026. Team: Detrix (Gabe Graves). Code: https://github.com/gabegraves/phase_eval (public, Apache-2.0). PhaseEval is the expert review, triage, scorecard and qualification product; AgentXRD is the analysis agent and harness it reviews.

The problem

AgentXRD is our XRD analysis agent. Given a powder pattern and the sample's declared precursors, a language model (here Qwen3.6-27B on our own GPU) searches a crystal structure library for candidate phases, runs Rietveld refinements in BGMN through recorded tool calls, and reports the phases, their amounts and the fit. A low fit statistic does not prove the phase list, so someone who knows diffraction has to look. PhaseEval shows each answer's checks, routes strong flags to an expert, supports blind judgment by unexposed raters, and turns eligible stored judgments into test cases that later models are graded against.

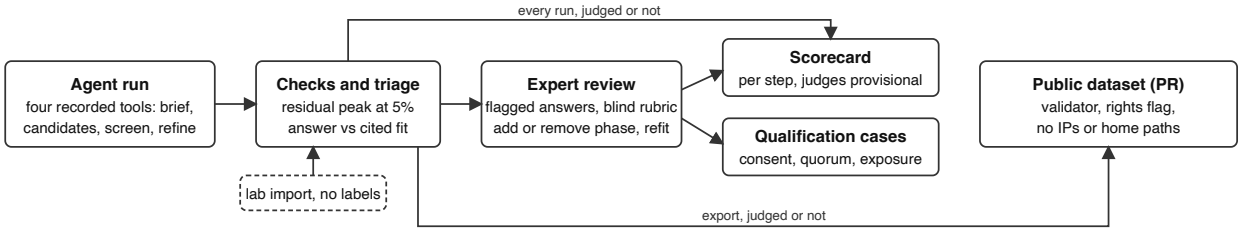


Figure 1. An agent run with four recorded tools gets checks and triage, and flagged answers go to expert review, whose judgments feed the scorecard and qualification cases. Any exported run, judged or not, can be sent as a pull request to the public dataset after the submit checks.

What we built

A triage policy reads each answer's checks (a residual peak at or above 5 percent of the strongest net peak, an answer that differs from its cited fit) and routes the answer to an expert; weaker peaks are listed for review. The threshold is provisional. The expert sees the evidence first: the fit with unexplained peaks flagged and chemistry evidence per phase (charge balance, hull distance, element balance). Then the agent's write-up, with record-backed checks on selected claims, and the rubric: is each phase supported by the pattern, chemically plausible for this sample, are the amounts defensible on their stated basis, does the fit support the conclusion. For unexposed raters the reference answer is withheld until the server has stored the sign-off. An expert can mark any refinement step, and can add or remove a phase and refit on BGMN, creating a child attempt beside its parent. Eligible human judgments become qualification cases under the configured review rules (consent, quorum, exposure; unresolved judgments stay out). A model's answers are graded against those cases with accepted and rejected rates kept apart. Qualification is shown only in tests on synthetic fixtures. Two model judges, GPT-6 Astra and Claude Opus, isolated and tool-less, each ran once over the 18 stored answers. They agree on 81 of 86 comparable items, but Opus said yes on 146 of 148, so kappa is 0.42. A model judge alone can fail a step but can no longer pass one, and model-judge reward terms are marked provisional until a judge is calibrated against expert labels. No expert labels exist yet. A scorecard lists every agent run by workflow step and keeps demonstration pairs apart from anything eligible for review or training. Eligible records export as versioned JSON with published schemas and a validator.

Open harness and public dataset

The public repository holds the four analysis tools (case brief, candidate table, peak screen, staged BGMN refinement) and one example case, DEV006, built from three open COD precursor structures (NiO 1010093, Li_2CO_3 9008283, alpha-Bi_2O_3 1526458, CC0). From a fresh clone, run.sh exits 0 and the staged refit ends at Rietveld Rwp 3.52 percent. This is one public example, not a validation for other labs or patterns. One command runs any OpenAI-compatible chat model that supports tool calls with those tools on a case; the endpoint URL is never recorded. One model was tested this way, Qwen3.6-27B, 7 turns, answer submitted. A lab can import a run made this way as a user case with no reference answer and no labels, and export a dataset the repo validator accepts (0 problems). An imported run joins the review queue and shows on the scorecard. A submit script sends an export to the public Hugging Face dataset kineticdrive/phaseeval-runs as a pull request for maintainer review; it refuses to send unless the submitter passes a rights flag, and refuses files carrying IP addresses or home paths. The dataset is seeded with the six public Qwen episodes.

The case in the video

DEV006 is a public DARA benchmark pattern (Zenodo 10.5281/zenodo.17410051, CC BY 4.0): the eight-minute scan of an unheated mix of NiO, Li_2CO_3 and Bi_2O_3. The video shows an episode by Qwen3.6-27B running locally: the three precursors and a residual peak near 24 degrees at about 3 percent of the strongest net peak. The check lists it as a weak peak for review, below the provisional flag. The labels in the video are revealed demonstration labels, not blind, and are excluded from evaluation. The published reference lists only the three precursors. DARA's own first solution had reported Bi_2CO_5, and a second model, GPT-6 Astra, added it at about 2.7 percent of modeled crystalline mass. Whether Bi_2CO_5 is present remains unresolved; the demonstration shows the review and refit tools, not expert adjudication.

From failure to fix, and next. When a run fails, PhaseEval writes a failure report for it: each step's outcome, what decided it, the evidence and the reviewer's reason, as a readable page and as JSON. An expert or another AI agent can use that report to fix the agent's tools or retrain the model, and the same cases can then grade the fix. **Coming very soon:** one-click setup of PhaseEval as a Prime Intellect environment, with reinforcement learning training runs directly on the platform, so the cases that grade an agent can also train it. A first version passes its tests on synthetic cases in our private code; it is not public yet, and no model has been trained with it.

What we do not claim

Before the event: the BGMN backend, the .xrdml reader, the agent tool routes and the harness. During the event: the rest of the work above, from the recorded episodes and review workbench to the public harness example and the dataset submission. Not public: the RL wrapper, the task builder, the candidate CIF library (some ICSD-derived) and the GSAS-II and Profex importers. No model was trained; we claim no learning gains, held-out accuracy or routing recall. Expert time is the scarce input. This spends it where it counts, and a lab can submit runs made with the public harness in one export format.